



PERSONALIZED CAREER PATH
SOLUTIONS

Career Path Recommendation System

Presented by

Marzia Tasneem (2410030446)

Chaitra Venugopal (2410030171)

Akshainie kumbaji(2410030506)



Career Guidance

With today's students facing increasing challenges in selecting suitable careers, it's crucial to provide **personalized guidance**. Current academic environments often lack tailored support, leading to uninformed decisions that can result in dissatisfaction and wasted resources. Our proposed system addresses these challenges by utilizing data analytics to offer informed career recommendations based on students' unique interests, skills, and academic backgrounds.



Objectives

The primary goals of the **Personalized Career Path Recommendation System** are to analyze student interests, skills, and academic backgrounds, providing tailored career path recommendations. Moreover, the system aims to suggest necessary skills and learning resources, ensuring a user-friendly experience. Integration of AI/ML technology will enhance the accuracy and adaptability of recommendations, ultimately empowering students in their career choices.



Current Limitations

Current career guidance methods often rely on static sessions with advisors or generic online questionnaires. These approaches lack **personalization** and fail to incorporate students' unique academic backgrounds, skills, and interests. Consequently, students are left with inadequate information, leading to uninformed decisions that can result in dissatisfaction and wasted resources. There is a pressing need for a more dynamic and intelligent guidance system.



Proposed System

Our full-stack web application leverages **AI technology** to analyze students' interests, skills, and academic backgrounds. By providing personalized career path recommendations, the system enhances decision-making and promotes informed choices. With an interactive dashboard and continuous updates based on user feedback, the application ensures that students receive tailored guidance, making their career exploration journey more efficient and engaging.



Methodology

The **methodology** for our system involves a systematic approach to data collection and processing. It begins with students inputting their interests, skills, and academic records. This data undergoes preprocessing, followed by AI/ML model processing that matches students to suitable career paths. The system then generates personalized recommendations and learning resources, continuously improving through a user feedback loop for enhanced accuracy and effectiveness.



Technologies Used in Our Project

1

Frontend

We utilize **React.js** to create a responsive and interactive user interface for seamless user experiences across devices.

2

Backend

The backend is powered by **Node.js** with Express for efficient server management and API interactions, ensuring smooth data flow.

3

Database

Our database solution is **MongoDB**, providing a flexible NoSQL structure to effectively store diverse student and career data.

Advantages

Our **Personalized Career Path System** offers significant benefits for students seeking tailored career guidance. By integrating diverse data sources, it delivers accurate, data-driven recommendations that adapt as users interact with the platform. This innovative solution not only enhances user satisfaction but also reduces career mismatches, paving the way for students to confidently pursue fulfilling career paths aligned with their interests and skills.



Applications

The **Personalized Career Path**

Recommendation System serves as a transformative tool for students and educational institutions. Students can explore tailored career options based on their unique skills and interests, while institutions enhance their counseling services, ensuring students receive informed guidance. Career advisors benefit from data-driven recommendations, helping them support students more effectively and prepare potential candidates aligned with industry needs.





Future Improvements

Our system's future scope includes **incorporating real-time job market trends**, enhancing the user experience through a mobile app version for easy access, and integrating with online course platforms for direct enrollment opportunities. Additionally, we aim to enhance AI models with deep learning and natural language processing capabilities, and provide multi-language support to reach a broader audience and meet diverse student needs.

Empowering Decisions

Our full-stack web application revolutionizes how students approach career choices. By integrating interests, skills, and academic backgrounds with **AI-powered recommendations**, we provide a personalized experience. This innovative tool addresses existing limitations in career guidance, ultimately enhancing student satisfaction and success. It invites educational institutions to embrace a data-driven approach, fostering informed career decisions for a brighter future.



THANK YOU